# SIMPLE LINEAR REGRESSION

ST331

(Statistical Computing 3)

# SIMPLE LINEAR REGRESSION

- Simple linear regression models describe the effect that a particular variable, called the explanatory variable, might have on the value of a continuous outcome variable, called the response variable.
- The explanatory variable may be continuous, discrete, or categorical

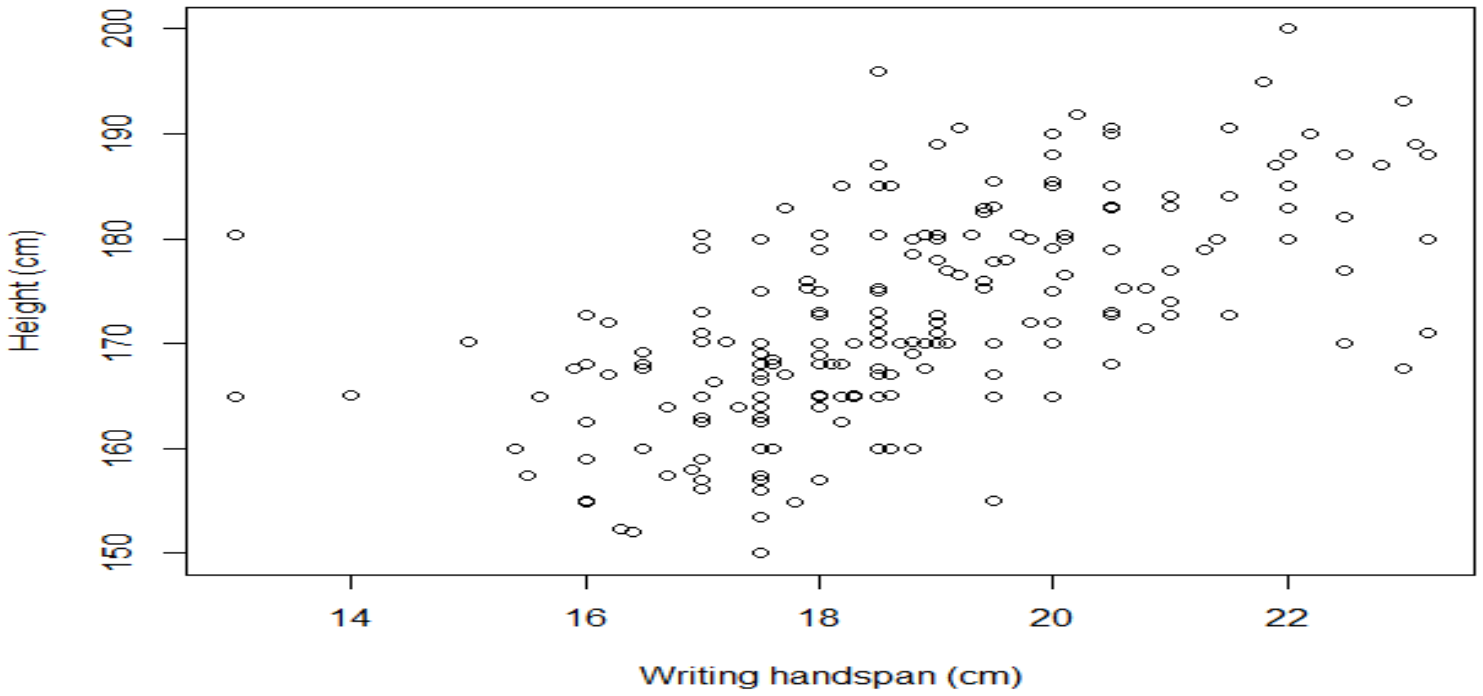
# SIMPLE LINEAR REGRESSION

As an example to start with, let's look at the student survey data (the survey data frame in the package MASS) a little more closely.

```
> library("MASS")
> data()
> ?survey
> survey
      Sex Wr.Hnd NW.Hnd W.Hnd    Fold Pulse    Clap Exer Smoke Height      M.I    Age
1  Female   18.5   18.0 Right  R on L    92    Left Some Never 173.00   Metric 18.250
2    Male   19.5   20.5  Left  R on L   104    Left None Regul 177.80 Imperial 17.583
3    Male   18.0   13.3 Right  L on R    87 Neither None Occas     NA     <NA> 16.917
4    Male   18.8   18.9 Right  R on L    NA Neither None Never 160.00   Metric 20.333
5    Male   20.0   20.0 Right Neither    35   Right Some Never 165.00   Metric 23.667
```

# SIMPLE LINEAR REGRESSION

```
plot(survey$Height~survey$Wr.Hnd,xlab="Writing handspan (cm)",ylab="Height (cm)")
```



# SIMPLE LINEAR REGRESSION

To assess the strength of the linear relationship, the estimated correlation coefficient is given as:

```
> cor(survey$Wr.Hnd,survey$Height,use="complete.obs")
[1] 0.6009909
```

In R, the command lm performs the estimation for you. For example, the following line creates a fitted linear model object of the mean student height

```
> survfit <- lm(Height~Wr.Hnd,data=survey)
> survfit

Call:
lm(formula = Height ~ Wr.Hnd, data = survey)

Coefficients:
(Intercept)       Wr.Hnd
    113.954        3.117
```

# SIMPLE LINEAR REGRESSION

```
> summary(survfit)

Call:
lm(formula = Height ~ Wr.Hnd, data = survey)

Residuals:
     Min       1Q   Median       3Q      Max
-19.7276  -5.0706  -0.8269   4.9473  25.8704

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 113.9536     5.4416   20.94   <2e-16 ***
Wr.Hnd        3.1166     0.2888   10.79   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.909 on 206 degrees of freedom
  (29 observations deleted due to missingness)
Multiple R-squared:  0.3612,    Adjusted R-squared:  0.3581
F-statistic: 116.5 on 1 and 206 DF,  p-value: < 2.2e-16
```

# REGRESSION COEFFICIENT SIGNIFICANCE TESTS

With a null value of zero, truth of H0 implies that the predictor has no effect on the response

$$H_0 : \beta_j = 0$$
$$H_\Lambda : \beta_j \neq 0$$

```
> confint(survfit,level=0.95)
                  2.5 %     97.5 %
(Intercept) 103.225178 124.682069
Wr.Hnd        2.547273   3.685961
```

This indicates that you should be 95 percent confident the true value of β1 lies somewhere between 2.55 and 3.69 (to 2 d.p.). The exclusion of the null value of zero reflects a statistically significant result.

# COEFFICIENT OF DETERMINATION

```
> rho.xy <- cor(survey$Wr.Hnd,survey$Height,use="complete.obs")
> rho.xy
[1] 0.6009909
> rho.xy^2
[1] 0.3611901
```

# OTHER SUMMARY OUTPUT

```
> names(summary(survfit))
 [1] "call"         "terms"        "residuals"    "coefficients"  "aliased"      "sigma"        "df"           "r.squared"    "adj.r.squared"
[10] "fstatistic"   "cov.unscaled" "na.action"
> summary(survfit)$r.squared
[1] 0.3611901
> summary(survfit)$adj.r.squared
[1] 0.3580891
> summary(survfit)$coefficients
             Estimate Std. Error  t value     Pr(>|t|)
(Intercept) 113.953623  5.4416389 20.94105 6.361804e-53
Wr.Hnd        3.116617  0.2887804 10.79234 8.227549e-22
```

# PREDICTION

let's say you want to determine the mean height for students with a handspan of 14.5 cm and for students with a handspan of 24 cm.

```
> 113.953623 +  3.116617* 14.5
[1] 159.1446
> 113.953623 +  3.116617* 24
[1] 188.7524
```

According to the model, you can expect mean heights to be around 159.14 and 188.75 cm for handspans of 14.5 and 24 cm, respectively

# MULTIPLE LINEAR REGRESSION

```
> demo.data <- data.frame(y=c(1.55,0.42,1.29,0.73,0.76,-1.09,1.41,-0.32),x1=c(1.13,-0.73,0.12,0.52,-0.54,-1.15,0.20,-1.09),x2=c(1,0,1,1,0,1,0,1))
> demo.data
     y    x1 x2
1 1.55  1.13  1
2 0.42 -0.73  0
3 1.29  0.12  1
4 0.73  0.52  1
5 0.76 -0.54  0
6 -1.09 -1.15  1
7 1.41  0.20  0
8 -0.32 -1.09  1
```

# MULTIPLE LINEAR REGRESSION

```
> demo.fit <- lm(y~x1+x2,data=demo.data)
> demo.fit

Call:
lm(formula = y ~ x1 + x2, data = demo.data)

Coefficients:
(Intercept)           x1           x2
      1.225        1.015       -0.698

> coef(demo.fit)
(Intercept)          x1          x2
  1.2254572   1.0153004  -0.6980189
```

# MULTIPLE LINEAR REGRESSION

```
> survmult <- lm(Height~Wr.Hnd+Sex,data=survey)
> survmult

Call:
lm(formula = Height ~ Wr.Hnd + Sex, data = survey)

Coefficients:
(Intercept)       Wr.Hnd      SexMale
    137.687        1.594        9.490
```

# MULTIPLE LINEAR REGRESSION

```
> summary(survmult)

Call:
lm(formula = Height ~ Wr.Hnd + Sex, data = survey)

Residuals:
     Min       1Q   Median       3Q      Max
-17.7479  -4.1830   0.7749   4.6665  21.9253

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 137.6870     5.7131  24.100  < 2e-16 ***
Wr.Hnd        1.5944     0.3229   4.937 1.64e-06 ***
SexMale       9.4898     1.2287   7.724 5.00e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.987 on 204 degrees of freedom
  (30 observations deleted due to missingness)
Multiple R-squared:  0.5062,    Adjusted R-squared:  0.5014
F-statistic: 104.6 on 2 and 204 DF,  p-value: < 2.2e-16
```

The fitted model itself can be thought of as
"Mean height" = 137.687 + 1.594 × "handspan" + 9.49 × "sex"

where "handspan" is the writing handspan supplied in centimeters and "sex" is supplied as either 1 (if male) or 0 (if female)

# MULTIPLE LINEAR REGRESSION

Because of this, you could this model as two equations. Here's the equation for female students:

"Mean height" = 137.687 + 1.594 × "handspan"

Here's the equation for male students:

"Mean height" = (137.687 + 9.4898) + 1.594 × "handspan"
= 147.177 + 1.594 × "handspan"

# MULTIPLE LINEAR REGRESSION

```
> survmult2 <- lm(Height~Wr.Hnd+Sex+Smoke,data=survey)
> summary(survmult2)
```

```
Call:
lm(formula = Height ~ Wr.Hnd + Sex + Smoke, data = survey)

Residuals:
     Min       1Q   Median       3Q      Max
-17.4869  -4.7617   0.7604   4.3691  22.1237

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 137.4056     6.5444  20.996  < 2e-16 ***
Wr.Hnd        1.6042     0.3301   4.860 2.36e-06 ***
SexMale       9.3979     1.2452   7.547 1.51e-12 ***
SmokeNever   -0.0442     2.3135  -0.019    0.985
SmokeOccas    1.5267     2.8694   0.532    0.595
SmokeRegul    0.9211     2.9290   0.314    0.753
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.023 on 201 degrees of freedom
  (30 observations deleted due to missingness)
Multiple R-squared:  0.5085,    Adjusted R-squared:  0.4962
F-statistic: 41.59 on 5 and 201 DF,  p-value: < 2.2e-16
```

# REGRESSION COEFFICIENT SIGNIFICANCE TESTS

```
> anova(survmult2)
Analysis of Variance Table

Response: Height
           Df Sum Sq Mean Sq  F value    Pr(>F)
Wr.Hnd      1 7298.7  7298.7 147.9720 < 2.2e-16 ***
Sex         1 2912.4  2912.4  59.0447 6.662e-13 ***
Smoke       3   44.9    15.0   0.3033     0.823
Residuals 201 9914.3    49.3
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

# REGRESSION COEFFICIENT SIGNIFICANCE TESTS

```
> confint(survmult2,level = 0.95)
                2.5 %     97.5 %
(Intercept) 124.5010442 150.310074
Wr.Hnd        0.9534078   2.255053
SexMale       6.9426040  11.853129
SmokeNever   -4.6061148   4.517705
SmokeOccas   -4.1312384   7.184710
SmokeRegul   -4.8543683   6.696525
```

Note that the Wr.Hnd and SexMale variables were shown to be statistically significant in the earlier model summary and that their 95 percent confidence levels do not include the null value of zero. On the other hand, all the coefficients for the dummy variables associated with the smoking frequency predictor are all non significant, and their confidence intervals clearly include zero. This reflects the fact that the smoking variable isn't, as a whole, considered statistically significant in this particular model.

# CLASS EXERCISE

In the MASS package, you'll find the data frame cats, which provides data on sex, body weight (in kilograms), and heart weight (in grams) for 144 household cats (see Venables and Ripley, 2002, for further details); you can read the documentation with a call to ?cats. Load the MASS package with a call to library("MASS"), and access the object directly by entering cats at the console prompt

Fit a least-squares multiple linear regression model using heart weight as the response variable and the other two variables as predictors, and view a model summary.

i. Write down the equation for the fitted model and interpret the estimated regression coefficients for body weight and sex. Are both statistically significant? What does this say about the relationship between the response and predictors?

(Statistical Computing 3)